

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CG

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INDEX NUMBER

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BIOLOGY

9744/01

Paper 1 Multiple Choice

13 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, CG and index number in the spaces at the top of this page.

Write in soft pencil.

Write your name, CG and index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

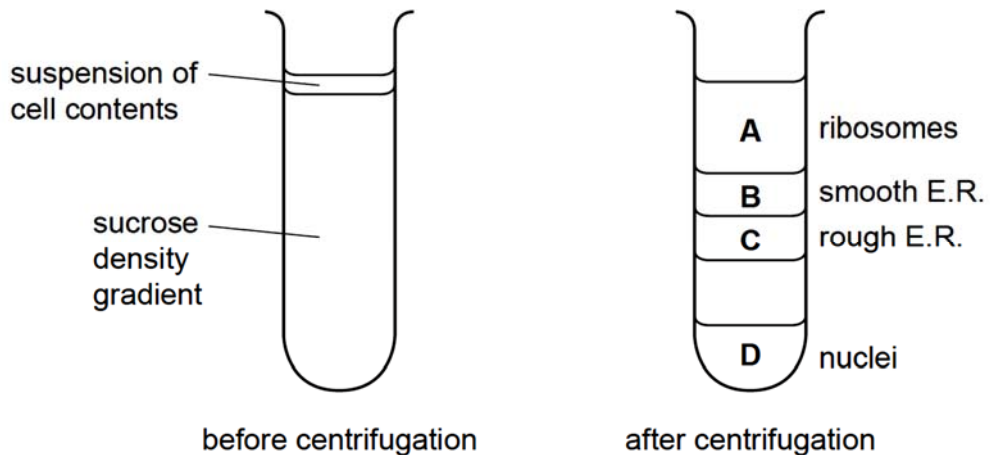
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **18** printed pages.



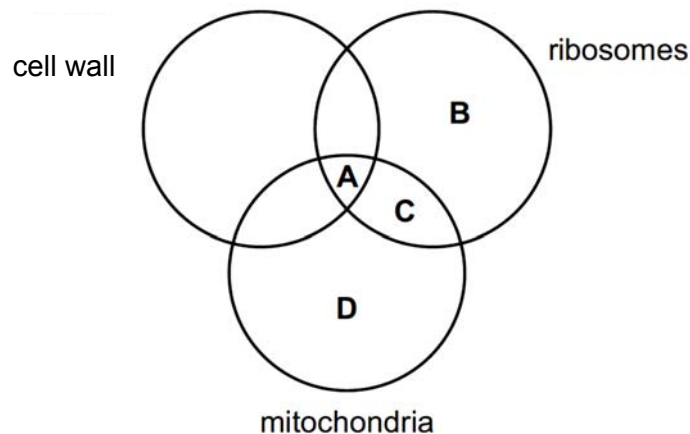
- 1 Sometimes scientists need to isolate organelles. This can be achieved by taking a number of cells and breaking their cell surface membranes to release the contents of the cells into a buffer solution.

In zonal centrifugation the suspension of cell contents is placed on top of a sucrose density gradient. The tube is then placed in a centrifuge and spun at high speed. The larger and denser particles will move towards the bottom of the tube faster than smaller and less dense particles as shown below.

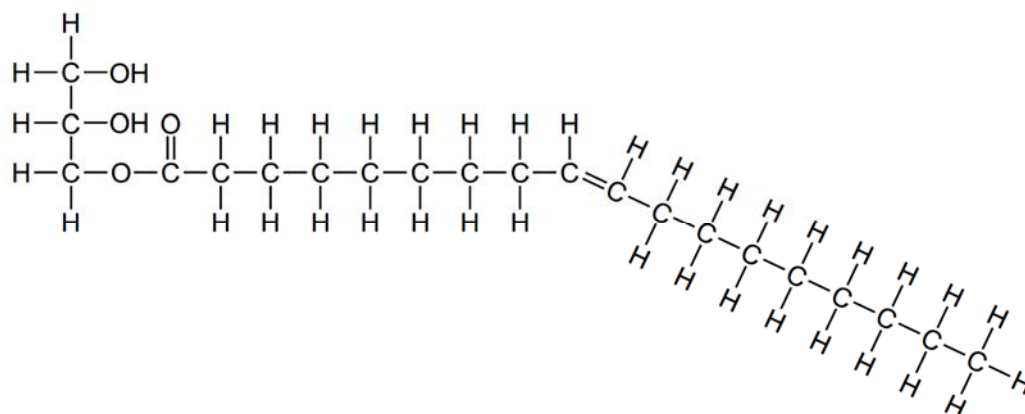


If a sample of intact prokaryotes had been added to a suspension of eukaryotic cell contents, where would you expect them to be found?

- 2 Which structures are present in a yeast cell?



- 3 The diagram shows a triglyceride molecule that has been partially hydrolysed.

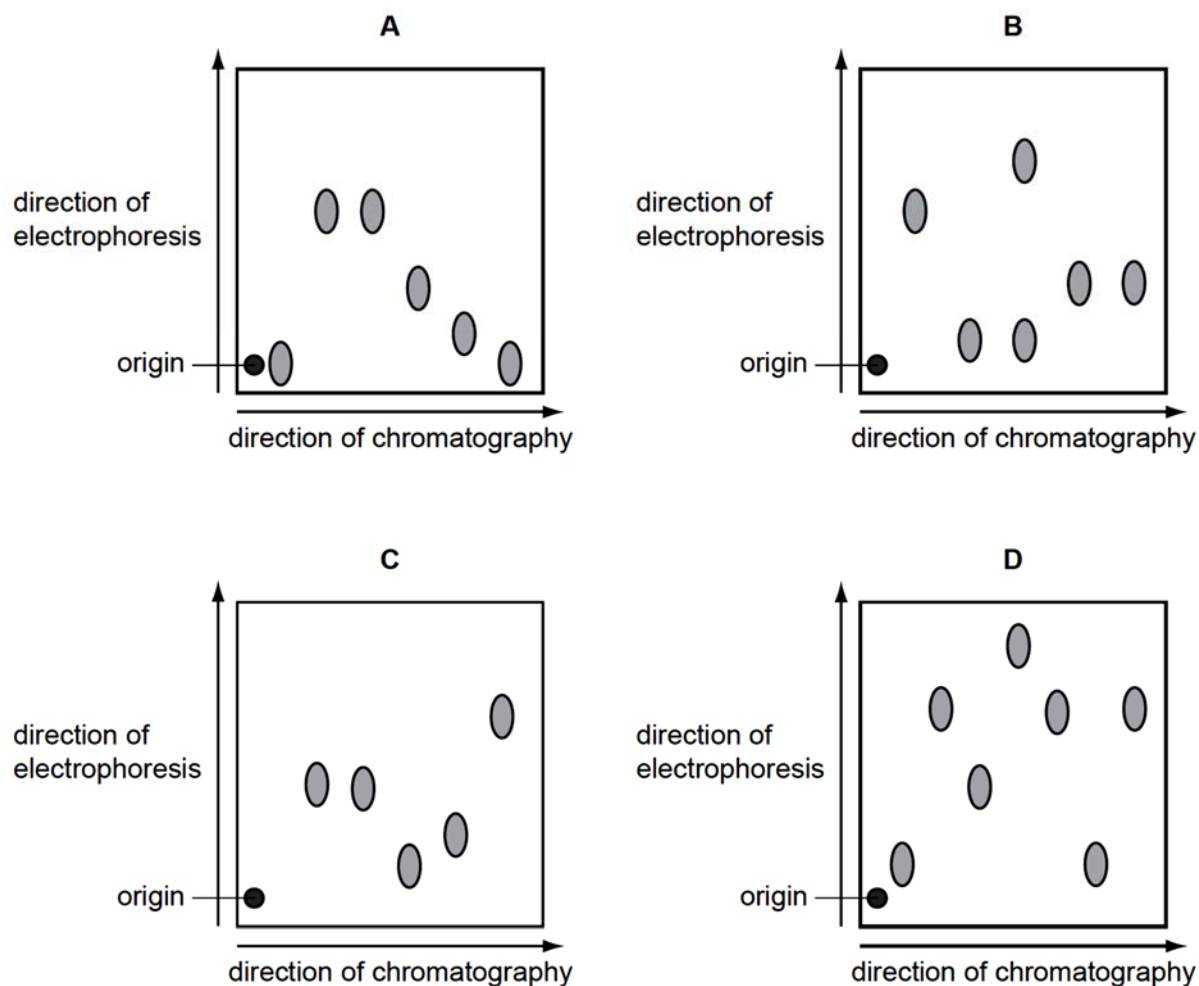


What will be the products of the total hydrolysis of the molecule shown?

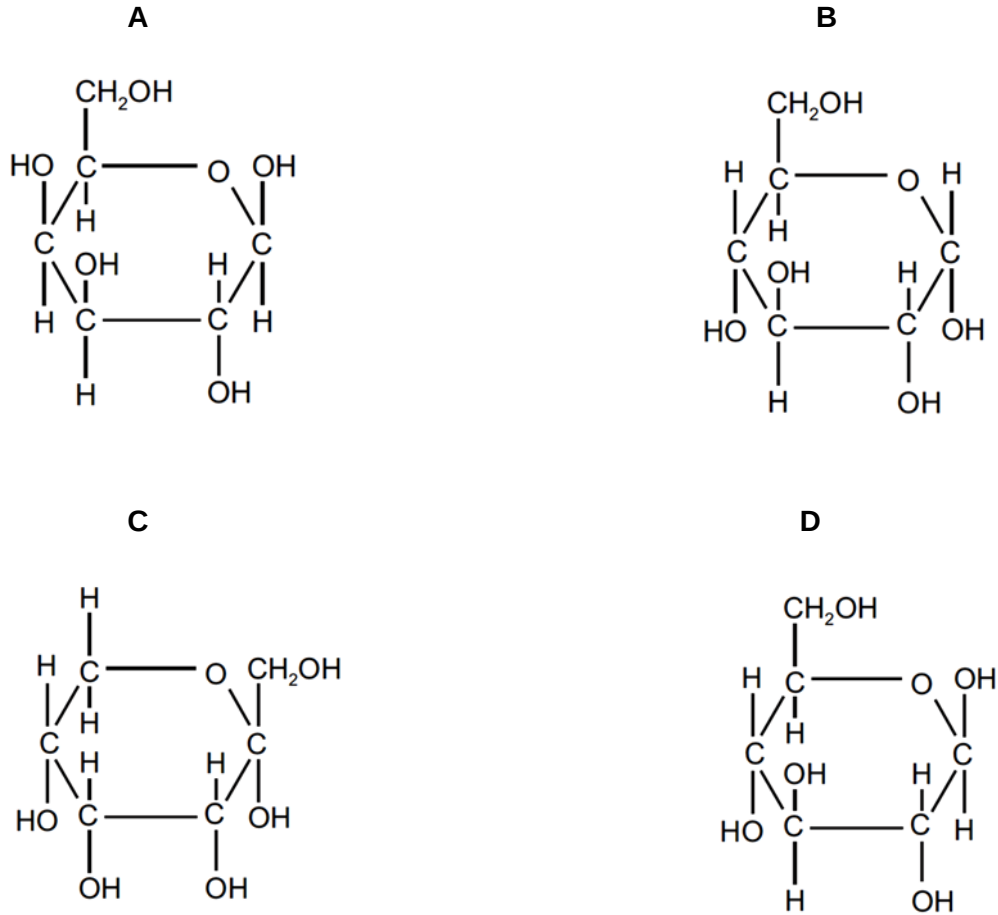
- A a molecule of glycerol and a saturated fatty acid molecule only
- B a molecule of glycerol and an unsaturated fatty acid molecule only
- C a molecule of glycerol and three fatty acid molecules
- D a molecule of water, a molecule of glycerol and a fatty acid molecule

- 4 The diagrams show the results of an investigation into the composition of different mixtures of amino acids. Each mixture of amino acids was separated using chromatography. Each chromatogram was then turned through 90° and the amino acids separated again by electrophoresis.

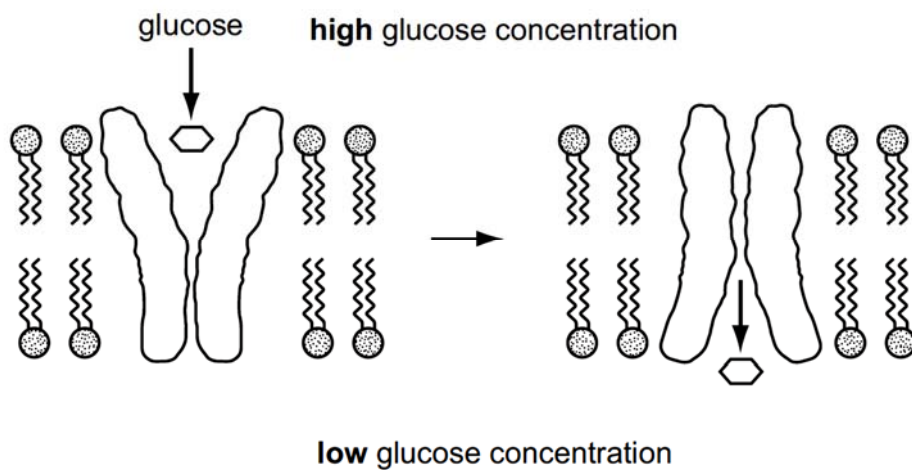
Which diagram shows an amino acid mixture in which the solubility of some of the amino acids is the same but the charge on those particular amino acids is different?



5 Which of the following shows a β -glucose?



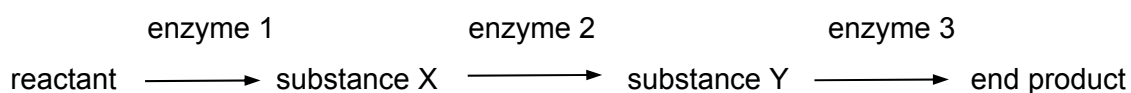
6 The diagram represents stages in glucose uptake through a cell surface membrane.



Which process is shown?

- A** active transport
- B** facilitated diffusion
- C** osmosis
- D** simple diffusion

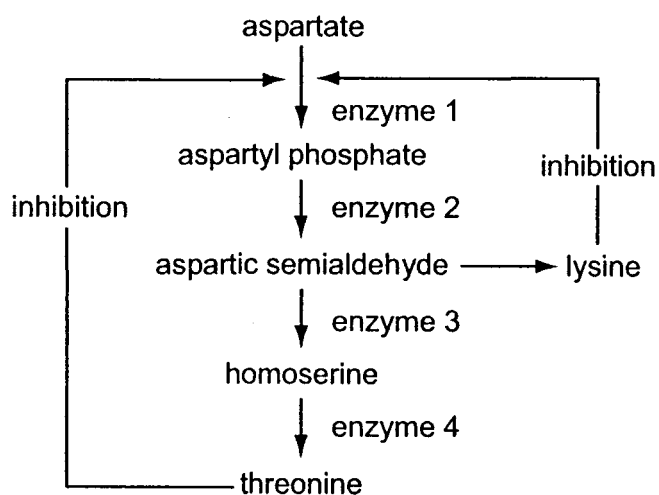
7 A metabolic pathway is



What would be the effect of adding a small amount of a non-competitive inhibitor of enzyme 2?

- A Enzyme 2 would be partially denatured.
- B Substance X would increase in concentration.
- C Substance Y would no longer be formed.
- D The initial reactant would no longer be metabolised.

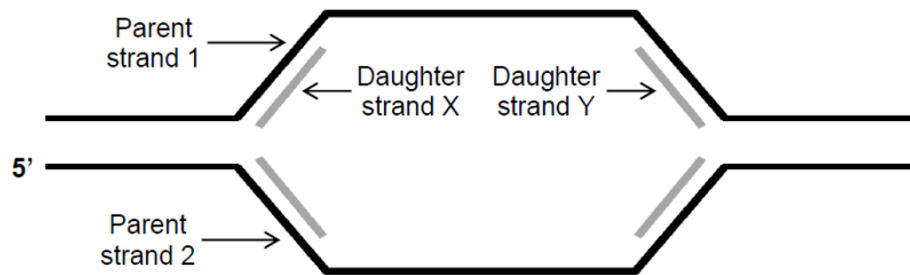
8 A culture of bacteria produces the food supplement lysine by the metabolic pathway shown.



Which change in enzyme activity will result in the greatest increase in lysine yield?

	enzyme	change in activity
A	1	decrease
B	2	increase
C	3	increase
D	4	decrease

- 9 A simplified representation of a replication bubble is shown in the figure below. Parental strands 1 and 2 and the growing daughter strands X and Y are indicated.

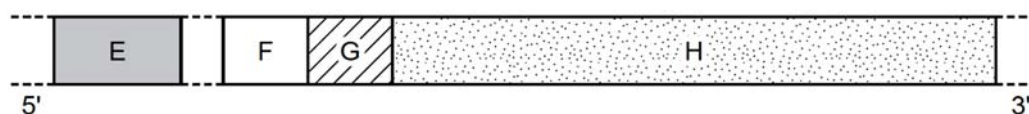


Which of the following statements about the syntheses of daughter strands X and Y is correct?

- A Daughter strands X and Y are synthesised away from their respective replication forks.
 - B Daughter strand X is synthesised continuously while daughter strand Y is synthesised in the form of Okazaki fragments.
 - C Daughter strand X is synthesised in the $5' \rightarrow 3'$ direction while daughter strand Y is synthesised in the $3' \rightarrow 5'$ direction.
 - D DNA ligase will eventually catalyse the fusion of daughter strand X with daughter strand Y.
- 10 Which of the following statement(s) is/are **not** true of the translation process in all eukaryotes?
- 1 Polypeptides are only synthesised in the cytosol.
 - 2 Amino acids are linked by the formation of peptide bonds catalysed by a ribozyme.
 - 3 Ribosomes contain an amino-acyl tRNA site that is occupied by the initiator tRNA attached to methionine.
 - 4 Amino-acyl tRNA synthetase attaches an amino acid to the 5' end of a tRNA molecule.
- A 1, 3 and 4 only
 - B 2, 3 and 4 only
 - C 2 and 4 only
 - D 1 only

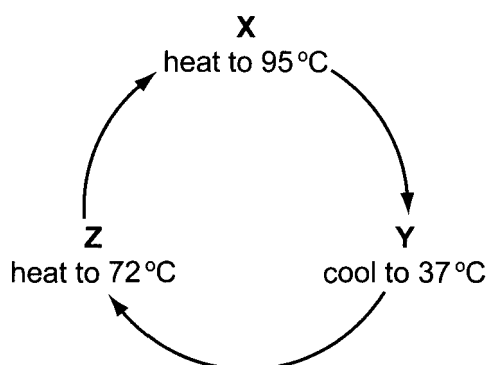
- 11** The diagram represents a length of DNA which forms a structure called an operon. Parts of the operon are labelled E, F, G and H. They have different functions.

What identifies the functions of parts E, F, G and H?



	E	F	G	H
A	operator	structural gene(s)	regulator/repressor	promoter
B	promoter	regulator/repressor	structural gene(s)	operator
C	regulator/repressor	promoter	operator	structural gene(s)
D	structural gene(s)	operator	promoter	regulator/repressor

- 12** The diagram shows the changes in temperature of the reaction mixture during the polymerase chain reaction (PCR).



The main events during one cycle of the reaction are listed.

- 1 binding of DNA primers
- 2 DNA synthesis
- 3 separation of DNA strands

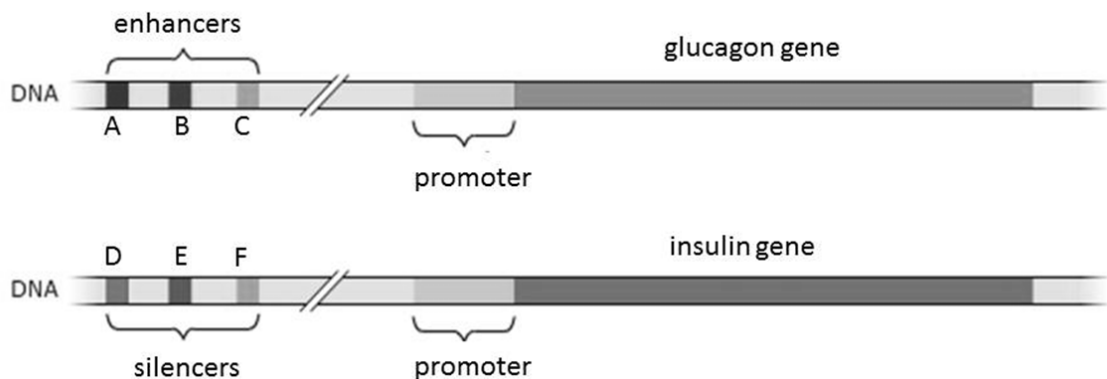
Which combination correctly matches each event with the temperature in the reaction mixture?

	X	Y	Z
A	1	2	3
B	2	3	1
C	3	1	2
D	3	2	1

13 Which combination correctly identifies the dengue virus?

	DNA	RNA	protein	phospholipid
A	✓	✓	✓	
B	✓		✓	✓
C		✓	✓	✓
D	✓		✓	

14 The diagram below shows the control elements and two genes found in the human genome.



Which of the following statement(s) about the above genes is/are true?

- 1 The glucagon gene is found only in the α -cells of the Islets of Langerhans while the insulin gene is found only in the β -cells of the Islets of Langerhans.
- 2 Binding of control elements, specific transcription factors and RNA polymerase at the promoter initiates transcription of glucagon.
- 3 The glucagon gene will be transcribed at a high level when transcription factors bind to control elements A, B, and C.
- 4 The expression of insulin can only be suppressed when transcription factors bind to control elements D, E and F.

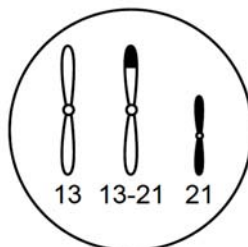
- A 3 only
 B 2 and 3 only
 C 1, 2 and 4 only
 D 2, 3 and 4 only

15 Which of the following is an example of translational control of gene expression?

- A The activation of proteins by association with other proteins
 B The addition of chemical groups such as phosphates to free amino acids
 C The binding of protein factors to mRNA to prevent the binding of the small ribosomal subunit
 D The degradation of a protein by proteasome

- 16** Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21.

The diagram shows chromosomes 13 and 21 in the nucleus of a diploid ($2n$) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



Which is **not** a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	$2n = 46$ normal phenotype
B	13-21	$2n = 45$ normal phenotype
C	13-21 and 21	$2n = 46$ Down's syndrome
D	13-21 and 21	$2n = 47$ Down's syndrome

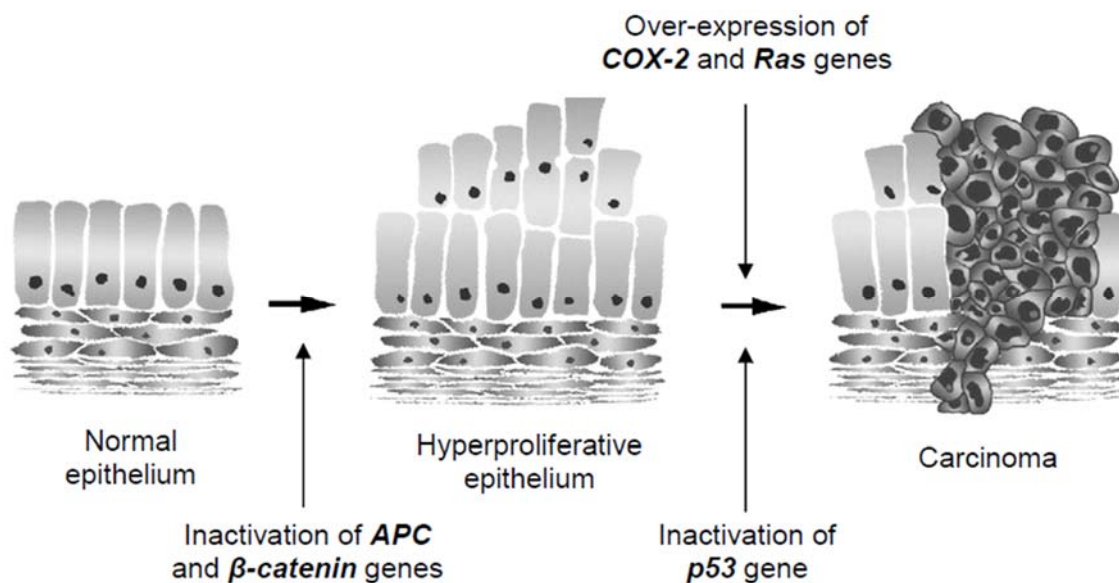
- 17** The diagram shows the chromosomes of one cell which has been squashed during mitosis.



Which stage of mitosis is shown and what is the haploid chromosome number in this species?

	stage of mitosis	haploid chromosome number
A	anaphase	5
B	anaphase	10
C	metaphase	5
D	metaphase	10

18 The diagram below illustrates the development of colorectal cancer.

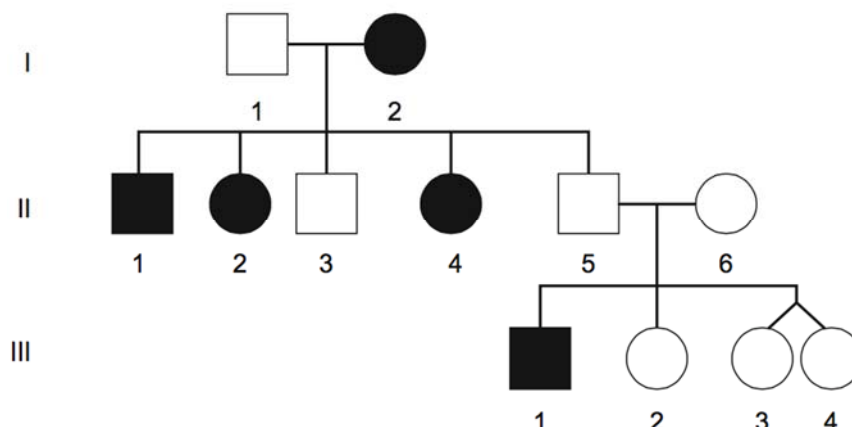


Which of these statements can be inferred from this multistep model of carcinogenesis?

- 1 Cells whose *APC* and *β-catenin* genes are inactivated have lost density dependent inhibition.
- 2 *APC* and *β-catenin* genes are most likely tumour suppressor genes.
- 3 High levels of *Ras* protein are produced only when both copies of *Ras* gene are mutated.
- 4 Two copies of normal *p53* alleles must be present to inhibit cell division.
- 5 Gain-of-function mutation in *COX-2* gene is one of the pre-requisites for the formation of carcinoma.

- A 1, 2 and 3
- B 1, 2 and 5
- C 2, 3 and 4
- D 2, 3 and 5

- 19 The pedigree below showed that the mode of inheritance of this disease is as supported by



- A autosomal dominant, individuals I-2 with offspring II-1, II-2 and II-4
 B autosomal recessive, individuals I-1, I-2 with offspring II-3 and II-5
 C autosomal recessive, individuals II-5, II-6 with offspring III-1
 D sex-linked dominant, offspring II-2 and II-4
- 20 The phenotypes of 200 offspring of a dihybrid test cross were recorded. The cross involved petal colour and fertility of the anthers of sweet pea flowers. The table shows the observed and expected numbers of each phenotype.

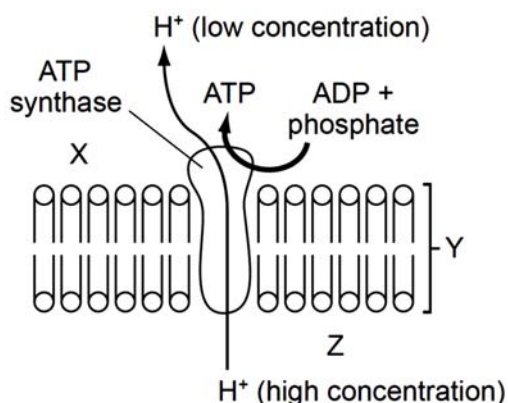
phenotype	purple petals fertile anthers	purple petals sterile anthers	maroon petals fertile anthers	maroon petals sterile anthers
observed numbers	87	14	16	83
expected numbers	50	50	50	50

A chi-squared (χ^2) test was performed and the probability of the difference between the observed and expected results being due to chance was found to be <0.001 .

Which conclusions may be drawn from this probability?

- 1 The difference is due to epistasis.
 - 2 The difference is due to chance.
 - 3 The difference is not due to chance.
 - 4 The difference is due to some factor such as linkage of the genes concerned.
- A 1 and 3 only
 B 1 and 4 only
 C 2 and 4 only
 D 3 and 4 only

- 21 The diagram shows a membrane in a cell.



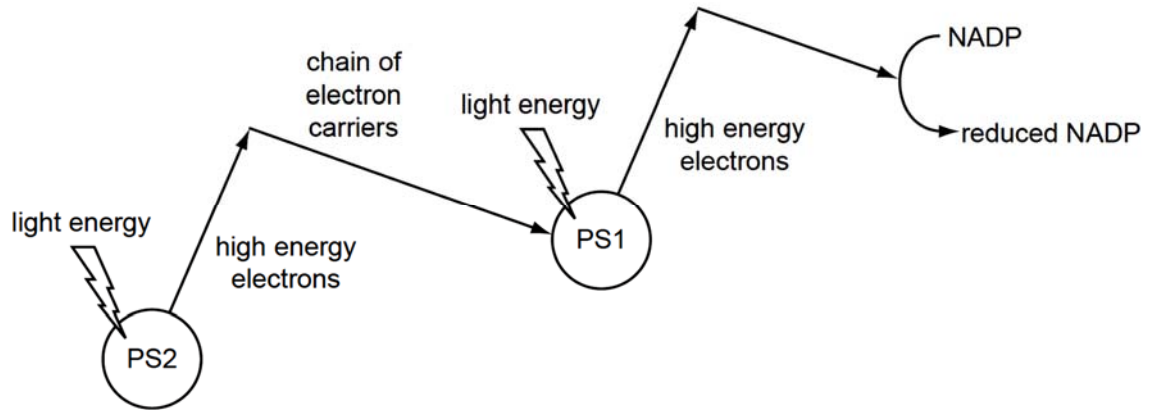
Which would be true of the diagram?

- A X is the thylakoid space, Y is the thylakoid membrane and the diagram shows ATP synthesis in a chloroplast.
 - B X is the stroma, Y is the thylakoid membrane and the diagram shows ATP synthesis in a mitochondrion.
 - C Y is the thylakoid membrane, Z is the cytosol (cytoplasm) and the diagram shows ATP synthesis in a chloroplast.
 - D Z is the intermembranal space, X is the matrix and the diagram shows ATP synthesis in a mitochondrion.
- 22 During substrate-level phosphorylation, ATP is synthesised from ADP and inorganic phosphate.

What is the immediate source of energy for this reaction?

- A chemical bond energy released during the light-independent stage of photosynthesis
- B chemical bond energy released during glycolysis and the Krebs cycle
- C kinetic energy of protons diffusing through mitochondrial membranes into the mitochondrial matrix
- D kinetic energy of protons diffusing through thylakoid membranes in chloroplasts

- 23 The diagram shows some of the processes in the light-dependent stage of photosynthesis.

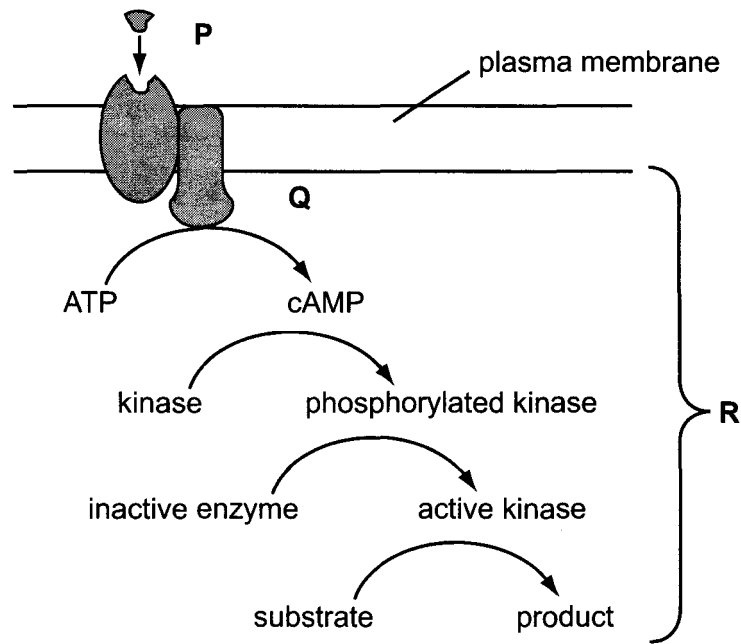


For the light-dependent stage to continue, photosystem two (PS2) must gain electrons.

Where do these electrons come from?

- A electron carriers
- B reduced NADP
- C photolysis of water
- D the formation of ATP

24 The diagram shows an example of cell signalling.



Which processes are shown on the diagram?

	P	Q	R
A	amplification	ligand-receptor interaction	phosphorylation
B	ligand-receptor interaction	amplification	phosphorylation
C	transduction	phosphorylation	amplification
D	ligand-receptor interaction	transduction	amplification

25 Which statements are true about Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species perfectly adapted to a stable environment will continue to evolve.
- 4 Variation between individuals of a species is essential for evolutionary change.

- A 1, 2 and 4 only
 B 2 and 3 only
 C 3 and 4 only
 D 4 only

- 26** Darwin's view of the process of evolution to form new species (speciation) has been reinforced by more recent discoveries in genetics and cell biology.

In this view, which sequence of events is considered most likely to lead to speciation?

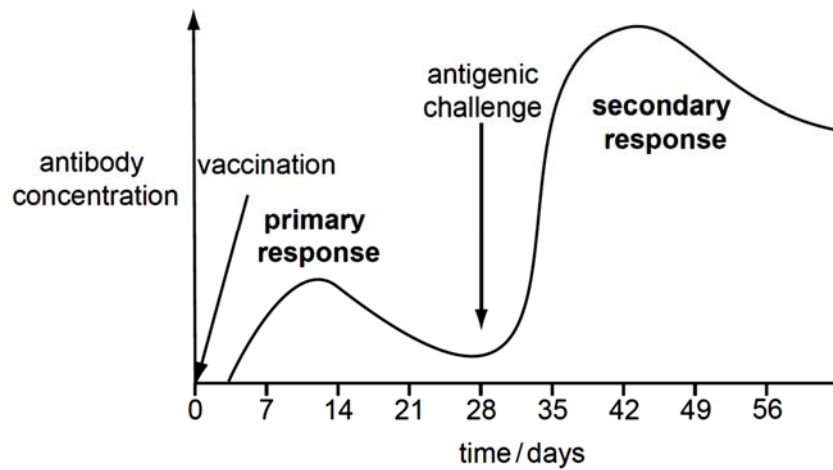
A	adaptation of population → competition and predation leading to natural selection → behavioural isolation → sympatric speciation
B	adaptation of population → competition and predation leading to natural selection → behavioural isolation → allopatric speciation
C	competition and predation leading to natural selection → geographical isolation → adaptation of isolated populations → sympatric speciation
D	competition and predation leading to natural selection → geographical isolation → adaptation of isolated populations → allopatric speciation

- 27** The huia, *Heteralocha acutirostris*, was found in New Zealand until 1907, when it became extinct. This bird had a ground-feeding habit and was particularly noted for large, attractive tail feathers. Males and females had very different beak forms, with the males having a short strong beak, whilst the females had a long curved beak to reach into otherwise inaccessible places.

What is the most likely reason for the extinction of the huia?

- A** Huia fed on species introduced by humans. When these declined, the huia population fell.
- B** In the face of a declining population the huia evolved into a tree-living species.
- C** Male and female huia were unable to breed successfully owing to strong sexual dimorphism.
- D** New competitors in New Zealand occupied part of the huia's niche.

- 28 The graph shows the level of antibody in serum following vaccination and a challenge with the same antigen 28 days later.

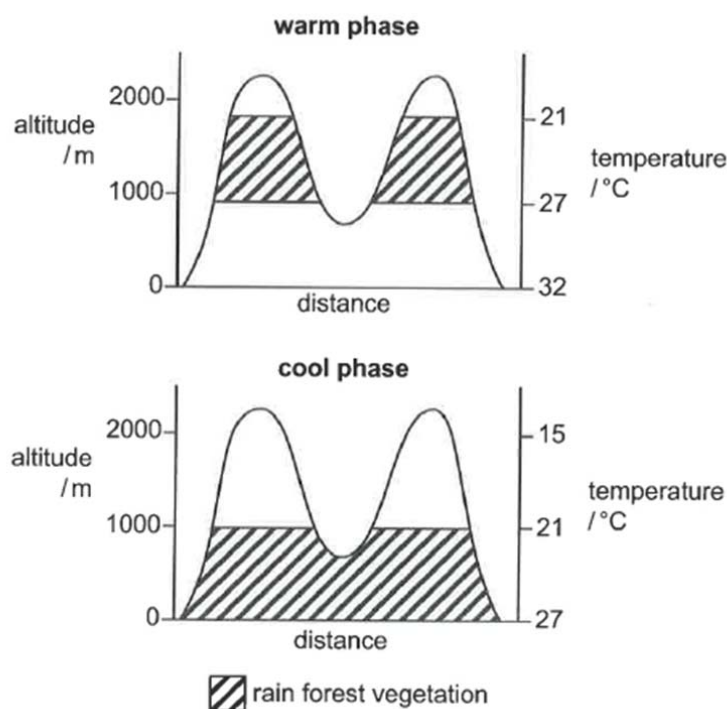


Which cells account for the difference in antibody concentration at the peaks of the primary and secondary responses?

- A B-lymphocytes
 - B memory cells
 - C phagocytes
 - D T-lymphocytes
- 29 What are the causative agents of AIDS, dengue and TB?

	AIDS	dengue	TB
A	human	bacterium	protozoan
B	bacterium	protozoan	virus
C	virus	mosquito	bacterium
D	virus	virus	bacterium

- 30** The diagram shows the topographical profile of two mountains in the tropics during natural warm and cool phases in the Earth's climate. The shape of the lines corresponds to a vertical section through the mountains to show their height and shape. The distribution of rain forest vegetation is also shown.



Which of the following statement is correct?

- A** The rain forest vegetation moves to stay in the suitable range of temperatures of 21 to 27°C.
- B** The lower altitudes are too hot during the warm phase in the Earth's climate for the growth of the rain forest vegetation.
- C** Climate change causes the rain forest vegetation distribution to increase in altitude with different temperature range.
- D** The changing rain forest vegetation distribution decreases evolution as selection pressure remains the same.